(b) Science and Technology

India has made great strides in developing institutions of scientific research and technology. The only science research institute in India before Independence was the Indian Institute of Science (IISc) established in 1909 in Bangalore with funding from J.R.D. Tata and the Maharaja of Mysore.



Indian Institute of Science, Bengaluru

The Tata Institute of Fundamental Research (TIFR) was set up in 1945 on the initiative of Homi J. Bhabha, with some funding from the Tatas. It was intended to promote research in mathematics and pure sciences. The National Chemical Laboratory, Pune and the National Physics Laboratory, New Delhi were the first institutes set up in India around the time of Independence. Since then there has been a steady increase in the number of institutes doing research in pure sciences, ranging from astrophysics, geology/ geo-physics, cellular and molecular biology, mathematical sciences and so on.

The Council of Scientific and Industrial Research (CSIR) is the umbrella organisation under which most of the scientific research institutions function. The CSIR also advances research in applied fields like machinery, drugs, planes etc.

The Atomic Energy Commission is the nodal agency for the development of nuclear science which is strategically important, focusing both on nuclear power generation and nuclear weapons. The Atomic Energy Commission also funds several institutes of pure science research.

Agriculture is another area where there has been a significant expansion of research and development. The Indian Council of Agricultural Research (ICAR) is the coordinating agency for the research done not only in basic agriculture, but also associated activities like fishery, forests, dairy, plant genetics, bio-technology, varieties of crops like rice, potato, tubers, fruits and pest control. Agricultural universities are also actively engaged in teaching and research on agricultural practices. There are 67 agricultural universities in India, and 3 in Tamil Nadu.



IIT, Chennai

Indian Institutes of Technology (IITs) were set up as centres of excellence in different fields of engineering. The first IIT was located in Kharagpur, followed by Delhi, Bombay, Kanpur and Madras (Chennai). There are now 23 IITs in the country, in addition to 31 NITs (National Institutes of Technology) and about 23 IIITs (Indian Institutes of Information Technology).

In spite of advances, the general perception is that science research in India still has a long way to go to catch up with the more developed countries and China. The research output in theoretical fields is rather disappointing and scanty in spite of the number of research institutions in the country.

12th_History_EM_Unit_9 Folder.indd 126 24-03-2020 10:02:42